

Product Requirements Document (PRD)

Mortgage & Loan Calculator Web App

Version 2.0 | With Enhanced Output, Download & Sharing Features

PART 1: COMPETITIVE RESEARCH & MARKET ANALYSIS

1.1 Top Competing Websites Analysis

Website	Traffic/Month	Key Strengths	Weaknesses
Bankrate.com	45M+	Comprehensive tools, editorial content	Cluttered UX, aggressive ads
NerdWallet.com	30M+	Clean UI, comparison tools	Limited international coverage
Mortgage Calculator.org	15M+	Simple, fast	Minimal content depth
Calculator.net	20M+	Multi-category calculators	Generic, no personalization
Zillow.com	60M+	Real estate integration	Mortgage-focused only
MoneySavingExpert.com	18M+	UK-focused, trusted editorial	UK only
SmartAsset.com	10M+	Financial advisor matching	US-centric
HSH.com	5M+	Rate data, deep mortgage tools	Dated UX
Dave Ramsey	12M+	Debt-free philosophy tools	US-centric, opinionated
MyFICO.com	8M+	Credit score integration	Limited calculator variety

1.2 Features Found Across Top Competitors

Universal Features (Must-Have)

- Basic mortgage payment calculator
- Amortization schedules (monthly/yearly)
- Refinance calculators
- Affordability calculators
- Interest rate comparison
- Printable/downloadable results (PDF)
- Email results functionality

Advanced Features (Differentiators)

- Real-time rate feeds
- Side-by-side loan comparisons
- Bi-weekly vs monthly payment analysis
- Extra payment impact calculators
- Tax deduction estimators
- PMI calculators
- Closing cost estimators

Content Strategy (SEO Moat)

- Glossary of financial terms
- "How it works" educational articles
- FAQ sections per calculator
- State/country-specific guides
- Blog with mortgage news/rate updates

PART 2: MOAT STRATEGY & DIFFERENTIATORS

2.1 Unique Value Propositions (Your Competitive Moat)

MOAT #1: True Multi-Country Architecture

Most competitors are US or UK only. You will support:

- **Tier 1:** USA, UK, Canada, Australia, Germany, France, India, UAE
- **Tier 2:** Singapore, New Zealand, South Africa, Brazil, Japan, Netherlands
- **Tier 3:** 40+ additional countries with localized rules

What this means technically:

- Local tax laws baked in per country
- Local currency with live exchange rates
- Country-specific loan types (Help-to-Buy UK, FHA/VA USA, etc.)
- Localized compliance terminology
- Multi-language support (Phase 2)

MOAT #2: AI-Powered "Loan Scenario Engine"

No major competitor has this:

- Users input their financial situation
- AI generates 3-5 personalized loan scenarios
- Plain-English explanation of each scenario
- "What changes if..." dynamic slider
- Compare scenarios side by side

- Save scenarios to account
-

MOAT #3: Real-Time Rate Intelligence Dashboard

- Aggregate public rate data across countries
 - Show rate trends (7-day, 30-day, 90-day, 1-year)
 - Rate alert subscription (email/SMS)
 - "Is now a good time to buy?" indicator
 - Central bank rate tracker (Fed, ECB, BoE, RBI, etc.)
-

MOAT #4: Interactive "Loan Journey" Gamification

- Progress tracker showing where user is in loan journey
 - Achievement badges for financial milestones
 - "Financial Health Score" based on inputs
 - Personalized tips based on score
 - Share score on social media (viral loop)
-

MOAT #5: Embeddable Widget Ecosystem

- Offer free embeddable calculators for real estate agents, brokers, financial blogs
 - Each widget backlinks to your site (massive SEO link-building moat)
 - White-label option for premium partners
 - API access for developers
-

MOAT #6: Multi-Discipline Calculator Breadth

Competitors focus on mortgages. You cover:

- Mortgage (all types)
 - Personal loans
 - Auto/vehicle loans
 - Student loans
 - Business loans
 - HELOC
 - Reverse mortgage
 - Construction loans
 - Bridge loans
 - Islamic finance (Halal loans - huge underserved market)
-

MOAT #7: Superior Output, Download & Sharing System

This is a PRIMARY differentiator. Most competitors offer basic PDF only. You will offer:

- **Multi-format downloads:** PDF, Excel, CSV, PNG image, JSON
- **Branded shareable reports:** Professional cover page, charts included

- **Social sharing:** Pre-formatted cards for LinkedIn, Twitter, WhatsApp
- **Shareable links:** Unique URL that recreates exact calculation
- **QR code generation:** For physical sharing with agents/advisors
- **Email delivery:** Send full report to self or advisor
- **Collaborative sharing:** Share with spouse/partner to edit together
- **Embed results:** Paste results into any website or forum
- **Print-optimized views:** Professional print layouts

MOAT #8: Dynamic SEO Content Engine

- Auto-generated location-specific pages
- Rate comparison pages by lender (affiliate revenue)
- Monthly "Rate Update" reports (email list building)
- User-submitted Q&A (UGC for SEO)
- Schema markup for rich snippets

MOAT #9: User Account & Dashboard

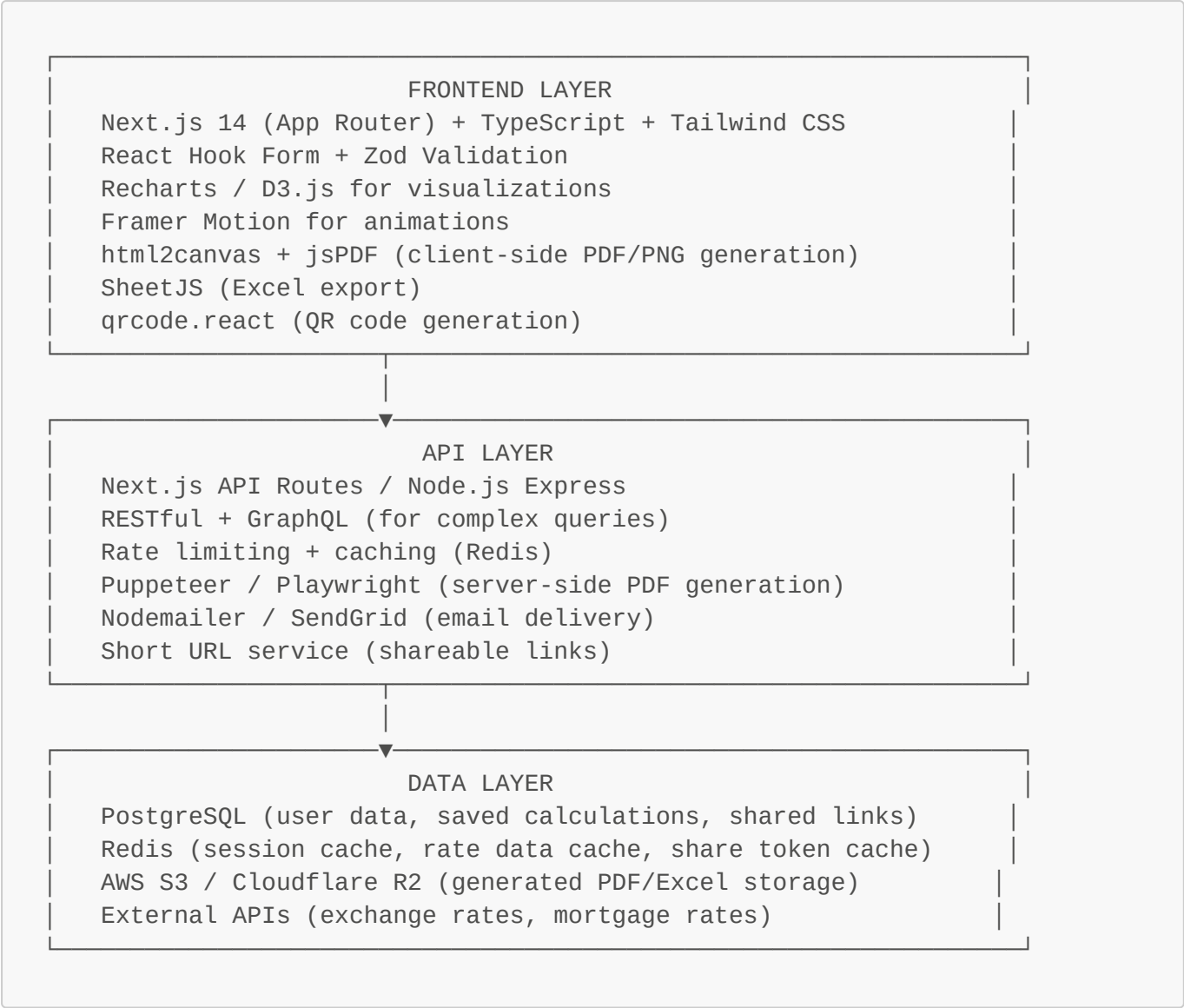
- Save unlimited calculations
- Track loan applications across time
- Set payment reminders
- Net worth tracker
- Document vault (Phase 2)

PART 3: FULL PRODUCT REQUIREMENTS DOCUMENT

3.1 Product Overview

Field	Detail
Product Name	CalcWise (working title)
Product Type	Web Application (SaaS + AdSense)
Primary Goal	Free-to-use financial calculator platform with AdSense monetization
Secondary Goal	Affiliate revenue, lead generation, premium subscriptions
Target Users	First-time homebuyers, refinancers, investors, financial advisors, real estate agents
Supported Countries	50+ (phased rollout)
Monetization	Google AdSense + Affiliate CPA + Premium Subscription

3.2 Technical Architecture



Tech Stack Decisions

Layer	Technology	Rationale
Framework	Next.js 14	SSR/SSG for SEO, App Router
Language	TypeScript	Type safety for financial calculations
Styling	Tailwind CSS + shadcn/ui	Speed + consistency
Charts	Recharts + D3.js	Complex financial visualizations
Client PDF	jsPDF + html2canvas	Instant client-side PDF
Server PDF	Puppeteer	High-fidelity server-rendered PDF
Excel Export	SheetJS (xlsx)	Full Excel with charts
CSV Export	Papa Parse	Clean CSV generation
Image Export	html2canvas	PNG snapshot of results

Layer	Technology	Rationale
QR Code	qrcode.react	QR for shareable links
Short URLs	Custom + nanoid	Shareable calculation URLs
Database	PostgreSQL + Prisma	Relational data, type-safe ORM
File Storage	AWS S3 / Cloudflare R2	PDF/Excel storage
Cache	Redis	Rate data, session, share tokens
Auth	NextAuth.js	OAuth + email auth
Email	Resend / SendGrid	Report delivery, alerts
Hosting	Vercel + AWS RDS	Edge performance
CDN	Cloudflare	Global speed, DDoS protection
Analytics	Google Analytics 4 + Mixpanel	Traffic + behavior
SEO	Next-SEO + Schema markup	Organic traffic

3.3 Site Architecture & URL Structure

```
calcwise.com/
├── / (Homepage)
├── /calculators/
│   ├── mortgage/
│   │   ├── basic-mortgage-calculator/
│   │   ├── affordability-calculator/
│   │   ├── refinance-calculator/
│   │   ├── amortization-schedule/
│   │   ├── extra-payment-calculator/
│   │   ├── bi-weekly-payment-calculator/
│   │   ├── arm-vs-fixed-calculator/
│   │   ├── fha-loan-calculator/
│   │   ├── va-loan-calculator/
│   │   ├── heloc-calculator/
│   │   ├── reverse-mortgage-calculator/
│   │   ├── construction-loan-calculator/
│   │   ├── bridge-loan-calculator/
│   │   ├── jumbo-loan-calculator/
│   │   ├── interest-only-calculator/
│   │   └── pmi-calculator/
│   ├── personal-loan/
│   ├── auto-loan/
│   ├── student-loan/
│   ├── business-loan/
│   └── islamic-finance/
```

```
|
|— /countries/
|   |— us/ | uk/ | canada/ | australia/
|   |— germany/ | india/ | uae/
|   |— [country-slug]/
|
|— /share/
|   |— [unique-token]/ ← Shareable calculation links
|
|— /rates/
|— /tools/
|   |— loan-comparison/
|   |— financial-health-score/
|   |— scenario-planner/
|
|— /learn/
|
|— /embed/
|   |— [calculator-id]/
|
|— /dashboard/ (authenticated)
|   |— saved-calculations/
|   |— shared-calculations/ ← NEW
|   |— download-history/ ← NEW
|   |— rate-alerts/
|   |— profile/
|
|— /api/
|   |— /calculate/
|   |— /rates/
|   |— /export/ ← NEW
|   |   |— /pdf/
|   |   |— /excel/
|   |   |— /csv/
|   |   |— /image/
|   |— /share/ ← NEW
|   |   |— /create/
|   |   |— /resolve/[token]/
|   |   |— /email/
|   |— /qr/ ← NEW
```

3.4 Calculator Specifications

3.4.1 Core Mortgage Calculator

Inputs:

HOME PRICE	[\$ _____]
DOWN PAYMENT	[\$ ____] [____%]

LOAN TERM

[10/15/20/25/30 yr]

INTEREST RATE

[__%] [ Live]

LOAN TYPE

[Fixed/ARM/IO]

START DATE

[Month/Year]

▼ ADVANCED OPTIONS

Property Tax

[\$ __ /year]

Home Insurance

[\$ __ /year]

HOA Fees

[\$ __ /month]

PMI Rate

[__%] [auto-calc]

Extra Payment

[\$ __ /month]

Outputs:

MONTHLY PAYMENT BREAKDOWN

Principal & Interest:

\$1,847

Property Tax:

\$312

Home Insurance:

\$83

PMI:

\$95

HOA:

\$150

TOTAL MONTHLY:

\$2,487

[Donut Chart - Payment Breakdown]

[Bar Chart - Principal vs Interest over time]

[Line Chart - Remaining Balance over time]

LOAN SUMMARY

Total Loan Amount:

\$380,000

Total Interest Paid:

\$284,491

Total Cost of Loan:

\$664,491

Payoff Date:

June 2055

OUTPUT ACTIONS TOOLBAR

[ PDF]

[ Excel]

[ CSV]

[ PNG]

[ Share Link]

[ Email]

[ QR Code]

[ Print]

[ WhatsApp]

[ Twitter]

[ LinkedIn]

[ Save to Account]

[AMORTIZATION SCHEDULE TABLE]

Year	Principal	Interest	Balance
2025	\$8,432	\$13,732	\$371,568
2026	\$8,943	\$13,221	\$362,625

Calculation Logic (TypeScript):

```

interface MortgageInputs {
  homePrice: number;
  downPayment: number;
  loanTermYears: number;
  annualInterestRate: number;
  propertyTaxAnnual: number;
  homeInsuranceAnnual: number;
  hoaMonthly: number;
  pmiRate: number;
  extraPaymentMonthly: number;
  loanType: "fixed" | "arm" | "interest-only";
}

interface MortgageOutputs {
  monthlyPrincipalInterest: number;
  monthlyPropertyTax: number;
  monthlyInsurance: number;
  monthlyPMI: number;
  monthlyHOA: number;
  totalMonthlyPayment: number;
  totalInterestPaid: number;
  totalCostOfLoan: number;
  payoffDate: Date;
  amortizationSchedule: AmortizationEntry[];
  loanAmount: number;
}

function calculateMortgage(inputs: MortgageInputs): MortgageOutputs {
  const {
    homePrice,
    downPayment,
    loanTermYears,
    annualInterestRate,
    extraPaymentMonthly,
  } = inputs;

  const loanAmount = homePrice - downPayment;
  const monthlyRate = annualInterestRate / 100 / 12;
  const numPayments = loanTermYears * 12;

  // Standard formula:  $M = P[r(1+r)^n]/[(1+r)^n-1]$ 
  const monthlyPI =
    (loanAmount * (monthlyRate * Math.pow(1 + monthlyRate,
numPayments))) /
    (Math.pow(1 + monthlyRate, numPayments) - 1);

  const schedule = generateAmortizationSchedule(

```

```
        loanAmount,
        monthlyRate,
        monthlyPI,
        extraPaymentMonthly,
    );

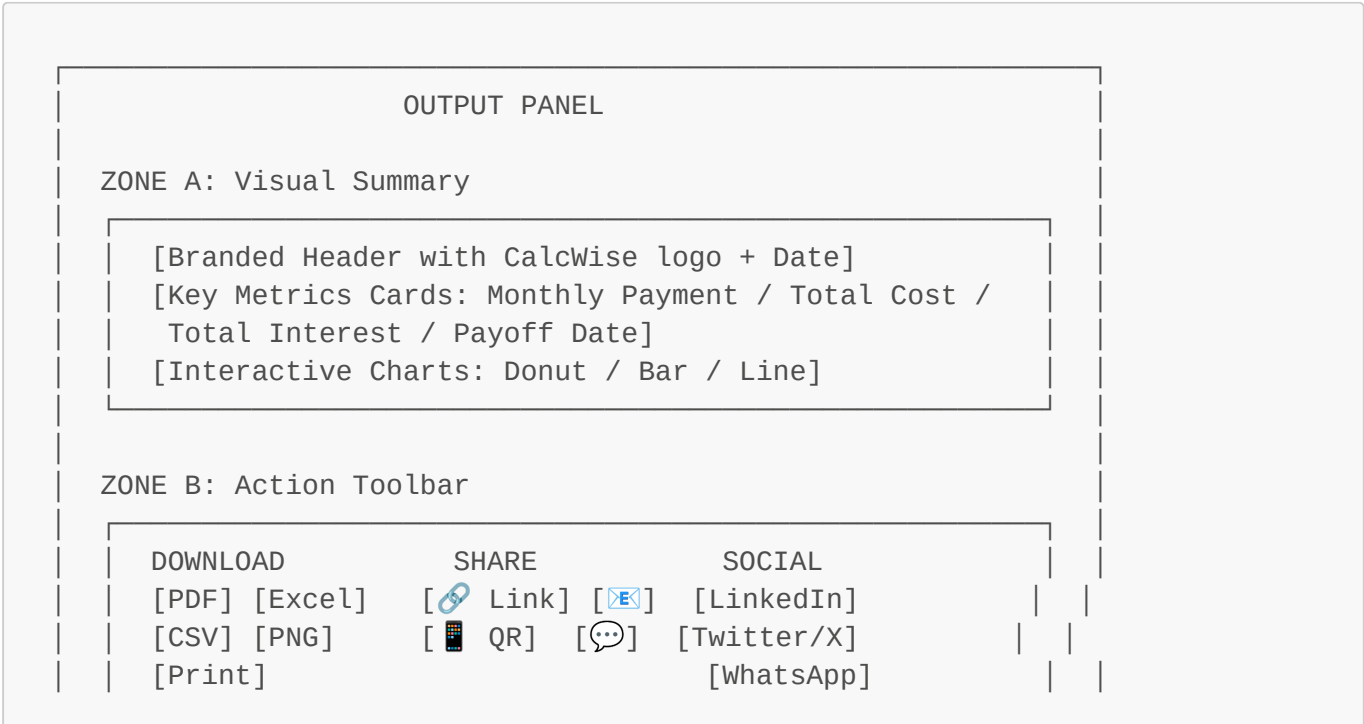
    return {
        monthlyPrincipalInterest: monthlyPI,
        monthlyPropertyTax: inputs.propertyTaxAnnual / 12,
        monthlyInsurance: inputs.homeInsuranceAnnual / 12,
        monthlyPMI: calculatePMI(loanAmount, homePrice, inputs.pmiRate),
        monthlyHOA: inputs.hoaMonthly,
        totalMonthlyPayment:
            monthlyPI +
            inputs.propertyTaxAnnual / 12 +
            inputs.homeInsuranceAnnual / 12 +
            inputs.hoaMonthly,
        totalInterestPaid: schedule.reduce((sum, e) => sum + e.interest,
0),

        totalCostOfLoan:
            loanAmount + schedule.reduce((sum, e) => sum + e.interest, 0),
        payoffDate: schedule[schedule.length - 1].date,
        amortizationSchedule: schedule,
        loanAmount,
    };
}
```

3.5 OUTPUT, DOWNLOAD & SHARING SYSTEM (NEW — CORE FEATURE)

3.5.1 Output Panel Architecture

The Output Panel is a persistent, sticky component that appears after calculation. It contains three zones:



ZONE C: Detailed Data

[Amortization Schedule - Full Table]
[Year-by-Year Summary]
[Payment Breakdown Detail]

3.5.2 Download Features (Detailed Specs)

 PDF Download

Two PDF Generation Modes:

Mode	Technology	Use Case	Quality
Quick PDF	jsPDF + html2canvas	Instant, client-side	Good
Full Report PDF	Puppeteer (server)	Professional, high-fidelity	Excellent

PDF Report Structure:

PAGE 1: COVER PAGE

[CalcWise Logo]

[Date Generated]

MORTGAGE CALCULATION REPORT

Prepared for: [User Name or "Guest User"]
Calculator: Basic Mortgage Calculator
Country: United States
Generated: January 15, 2025 at 3:42 PM EST
Reference #: CW-2025-011547392

KEY RESULTS AT A GLANCE

Monthly Payment	\$2,487
Total Interest	\$284,491
Total Cost	\$664,491
Payoff Date	June 2055
Loan Amount	\$380,000

Disclaimer: For informational purposes only.
Consult a licensed financial advisor.

PAGE 2: INPUT SUMMARY

YOUR CALCULATION INPUTS

Home Price: \$475,000
Down Payment: \$95,000 (20%)
Loan Amount: \$380,000
Loan Term: 30 Years
Interest Rate: 6.75% Fixed
Start Date: February 2025
Property Tax: \$3,750/year (\$312/mo)
Home Insurance: \$996/year (\$83/mo)
HOA Fees: \$150/month
PMI: N/A (≥20% down)
Extra Payment: \$0/month

PAGE 3: PAYMENT BREAKDOWN

[Donut Chart - Full Color]
[Monthly Payment Breakdown Bar Chart]
[Principal vs Interest Over Time Chart]

MONTHLY PAYMENT BREAKDOWN

Principal & Interest:	\$1,847	(74.3%)
Property Tax:	\$312	(12.5%)
Home Insurance:	\$83	(3.3%)
HOA Fees:	\$150	(6.0%)
PMI:	\$0	(0.0%)

TOTAL MONTHLY PAYMENT: \$2,487 (100.0%)

PAGE 4: AMORTIZATION SCHEDULE (YEARLY)

Year	Beginning Bal	Principal	Interest	End Bal
2025	\$380,000	\$8,432	\$13,732	\$371,568
2026	\$371,568	\$8,943	\$13,221	\$362,625
...				

PAGE 5: AMORTIZATION SCHEDULE (MONTHLY - FULL)

[Complete monthly breakdown]

PAGE 6: IMPORTANT NOTES & DISCLAIMER

[Legal disclaimer, data sources, advice to consult
licensed mortgage broker/financial advisor]
[Link to [calcwise.com](#) for recalculation]
[QR code to reload this exact calculation]



PDF DOWNLOAD OPTIONS

Report Type:

- Quick Summary (1-2 pages)
- Full Report (5-6 pages) [Recommended]

Include:

- ☒ Cover page
- ☒ Input summary
- ☒ Charts & visualizations
- ☒ Yearly amortization table
- ☒ Monthly amortization table
- ☒ Disclaimer page

Personalize (optional):

Your Name: [_____]

Note/Label: [_____]

Paper Size: [A4 ▼] [Letter] [Legal]

[CANCEL]

[↓ DOWNLOAD PDF]



Excel Download (.xlsx)

Excel File Structure:

WORKBOOK: CalcWise_Mortgage_2025-01-15.xlsx

Sheet 1: "Summary"

[CalcWise Logo] Mortgage Report
Generated: Jan 15, 2025

INPUT PARAMETERS

Home Price	\$475,000
Down Payment	\$95,000
Loan Amount	\$380,000
Interest Rate	6.75%
Loan Term	30 years

RESULTS

Monthly Payment	\$2,487
Total Interest	\$284,491
Total Cost	\$664,491
Payoff Date	June 2055

— Sheet 2: "Monthly Amortization"

Month	Date	Payment	Principal	Interest	Balance
1	Feb 2025	\$1,847	\$710	\$1,137	\$379,290
2	Mar 2025	\$1,847	\$713	\$1,134	\$378,577
... (360 rows)					

— Sheet 3: "Yearly Summary"

Year	Principal Paid	Interest Paid	Balance
2025	\$8,432	\$13,732	\$371,568
...			

— Sheet 4: "Payment Breakdown"

Category	Monthly	Annual	% of Payment
P&I	\$1,847	\$22,164	74.3%
...			

— Sheet 5: "Charts"

[Embedded Excel charts: Donut, Bar, Line]

Excel Features:

- ✓ Conditional formatting (red = high interest, green = principal growth)
- ✓ Frozen headers on scroll
- ✓ Auto-filter on amortization columns
- ✓ Color-coded milestone rows (25%, 50%, 75% paid off)
- ✓ Built-in Excel charts
- ✓ Print area pre-configured
- ✓ Custom CalcWise color theme
- ✓ Protected formula cells (prevent accidental editing)
- ✓ Data validation notes

 CSV Download

Format: calcwise_mortgage_2025-01-15.csv

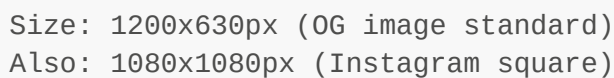
```
# CalcWise Mortgage Calculator Export
# Generated: 2025-01-15T15:42:00Z
# Calculator: Basic Mortgage Calculator
# Reference: CW-2025-011547392
#
# INPUTS
Home Price,$475000
Down Payment,$95000
Loan Amount,$380000
Interest Rate,6.75%
Loan Term,30 years
#
# MONTHLY AMORTIZATION
```

CSV Options:

 [PNG / Image Download](#)

Two Image Formats:

1. RESULTS CARD (Social Media Optimized)



2. FULL REPORT SCREENSHOT

Full-page render of results panel
High-resolution: 2x retina
Size: 800x[auto] px

 Print View

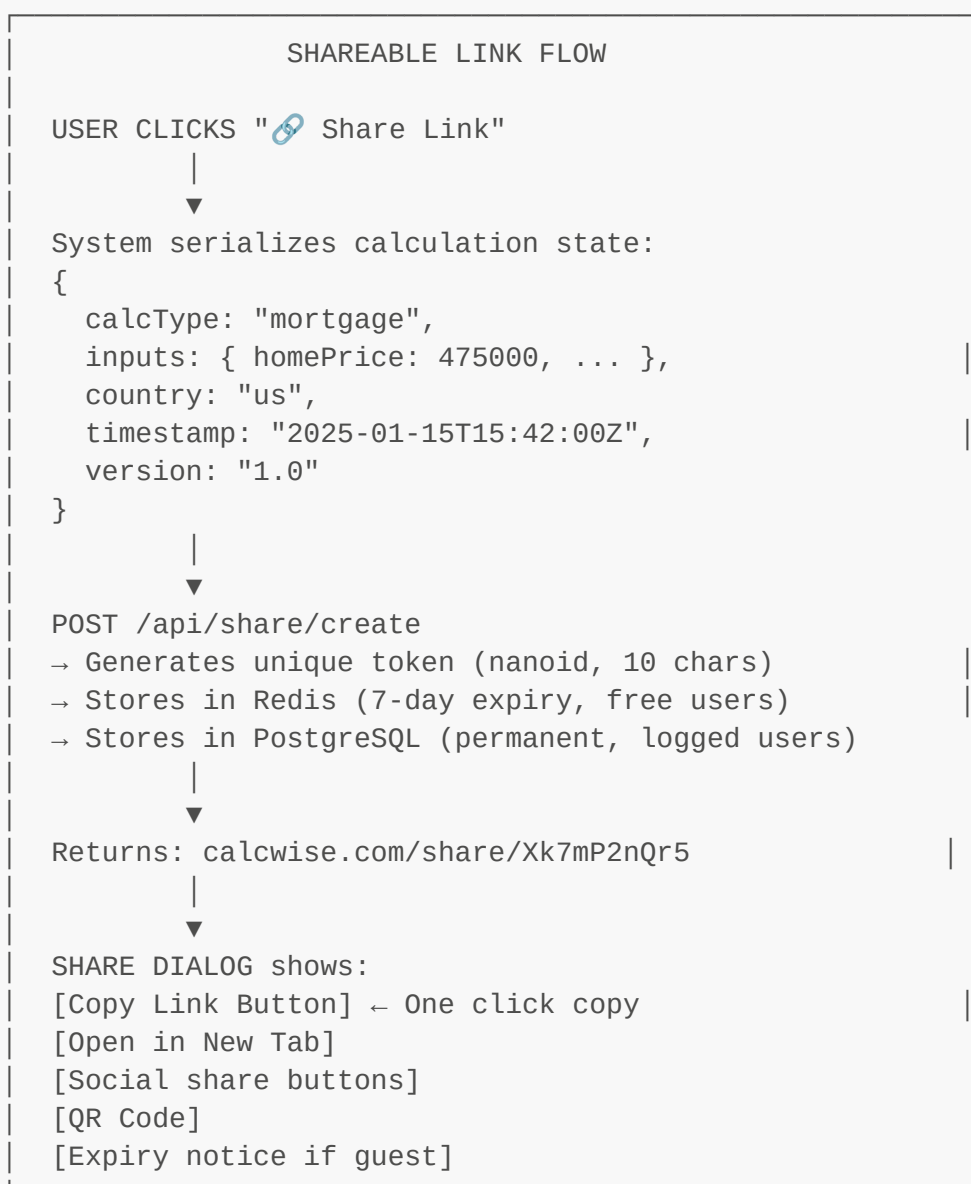
15 / 37

- ✓ Page break control (no table rows split)
- ✓ Show full URL for reference
- ✓ Include QR code for digital access
- ✓ Print-friendly font sizes
- ✓ Margin optimization for A4/Letter

Print Shortcut: Ctrl+P / Cmd+P triggers
optimized print view automatically

3.5.3 Shareable Link System

How It Works



Shareable Link Page (/share/[token])

When recipient opens link:

 BANNER: "Viewing a calculation shared by [Name]"
"Shared on Jan 15, 2025"
[Make My Own Copy] [Modify This Calc]

[FULL CALCULATOR shown with inputs pre-filled]
[Full results displayed]
[Download / Share options available to recipient]

SEO: noindex, nofollow (prevent duplicate content)

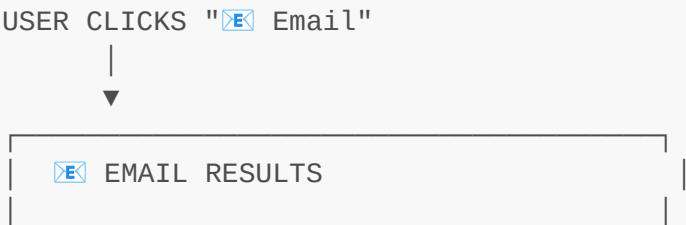
Share Link Management

```
interface ShareableLink {
  token: string; // "Xk7mP2nQr5"
  fullUrl: string; // "calcwise.com/share/Xk7mP2nQr5"
  calculationType: string;
  inputSnapshot: object; // Full inputs at time of share
  createdAt: Date;
  expiresAt: Date | null; // null = permanent (premium users)
  createdBy: string | null; // userId or null (guest)
  viewCount: number;
  isPasswordProtected: boolean;
  password?: string; // bcrypt hashed (premium feature)
  allowModification: boolean;
  allowDownload: boolean;
}

// Token link TTL rules:
// Guest users: 7 days
// Free account: 30 days
// Pro account: Permanent
// Business: Permanent + custom domain
```

3.5.4 Email Delivery System

Email Report Flow



Send to:
[your@email.com]

Add another recipient:
[advisor@email.com]
(e.g., your mortgage broker)

Subject line:
[My Mortgage Calculation - Jan 2025]

Personal message (optional):
[
]

Attach as:
☒ PDF Report ☒ Link to results
☐ Excel File ☐ CSV Data

[CANCEL] [ SEND EMAIL]

Email Template (HTML)

Subject: Your Mortgage Calculation Results - CalcWise [CalcWise Logo]
YOUR MORTGAGE CALCULATION RESULTS

Calculated on January 15, 2025 |
Monthly
Payment \$2,487/mo | | Total Interest \$284,491 | | Total Cost \$664,491 | |
Payoff
Date June 2055 | CALCULATION INPUTS:
Home Price: \$475,000 Down Payment:
\$95,000
(20%) Interest Rate: 6.75% Fixed Loan Term: 30 Years [VIEW FULL RESULTS
ONLINE
→] [Button: calcwise.com/share/Xk7mP2nQr5] [PIE CHART IMAGE EMBEDDED] Full
amortization schedule and detailed breakdown attached as PDF.

This is for informational purposes
only.
Consult a licensed financial professional. Unsubscribe | Privacy Policy
CalcWise.com

3.5.5 Social Sharing System

Platform-Specific Configurations

```
interface SocialShareConfig {  
    platform: string;
```

```

url: string;
text: string;
hashtags?: string[];
imageUrl?: string;
}

const socialShareConfigs = {
  twitter: {
    text: `Just calculated my mortgage with @CalcWise 🏠
Monthly Payment: $2,487/mo
Total Interest: $284,491
30-year fixed at 6.75%
Calculate yours FREE 📌`,
    hashtags: ["mortgage", "homebuying", "personalfinance"],
    url: "calcwise.com/share/Xk7mP2nQr5",
    characterLimit: 280,
  },

  linkedin: {
    text: `Planning a home purchase and used CalcWise to run the
numbers:

🏠 Home Price: $475,000
📊 Monthly Payment: $2,487/mo
💰 Total Interest Over 30 Years: $284,491
📅 Payoff Date: June 2055

The amortization breakdown was eye-opening.
Try the free calculator: calcwise.com`,
    type: "professional",
  },

  whatsapp: {
    text: `Hey! Check out this mortgage calculation I did:
🏠 Home: $475,000
💰 Monthly: $2,487
📊 Interest: $284,491 over 30 years
View full breakdown: calcwise.com/share/Xk7mP2nQr5`,
    isPersonal: true,
  },

  facebook: {
    url: "calcwise.com/share/Xk7mP2nQr5",
    // Uses OG meta image auto-generated per calculation
  },

  reddit: {
    title: "Used CalcWise to calculate my mortgage - sharing results",
    text: "Full breakdown including amortization schedule...",
    suggestedSubreddits: [
      "personalfinance",
      "FirstTimeHomeBuyer",
      "REBubble",
    ],
  },
}

```

```
},  
  
telegram: {  
  text: "My mortgage calculation from CalcWise 🏠",  
  url: "calcwise.com/share/Xk7mP2nQr5",  
},  
};
```

Social Share Modal UI

 SHARE YOUR RESULTS

Your shareable link:

[Copy]

✓ Link copied! Expires in 30 days

SHARE ON SOCIAL MEDIA:

 Twitter/X

 LinkedIn

 Facebook

 WhatsApp

 Telegram

 Reddit

 Email

 iMessage

SHARE VIA QR CODE:







Scan to open on mobile
or share with your agent
[Download QR Code]

EMBED THIS RESULT: (for website/forum)

<iframe src="calcwise.com/embed/share/
Xk7mP2nQr5" width="600" height="400">
</iframe>

[Copy]

 Link Settings

 Add Password (Pro)

3.5.6 QR Code System

```
interface QRCodeOptions {  
  url: string;  
  size: number; // 200x200 to 1000x1000
```

```

    errorCorrection: "L" | "M" | "Q" | "H";
    format: "PNG" | "SVG";
    includeLabel: boolean;
    labelText: string; // "Scan to view mortgage calculation"
    style: "standard" | "branded"; // branded = with CalcWise logo in
center
    foregroundColor: string;
    backgroundColor: string;
}

// QR Code Use Cases:
// 1. Share results with mortgage broker at meeting
// 2. Printed on PDF for quick digital access
// 3. Real estate agent sends to client
// 4. Workshop/seminar presenters

```

QR Code Download Options:

```

Size:    [Small 200px] [Medium 400px •] [Large 800px]
Format:  [PNG •] [SVG]
Style:   [Standard •] [With Logo (Pro)]
Color:   [Black/White •] [Custom (Pro)]
[Download QR Code]
[Print QR Code]

```

3.5.7 Collaborative Sharing (Premium Feature)

FEATURE: "Share & Edit Together"

User A calculates → Shares with User B (spouse/partner/advisor)

User B can:

- ✓ View all results
- ✓ Modify inputs (create own version)
- ✓ Add comments/annotations
- ✓ Compare User A vs User B scenarios side-by-side
- ✓ Both get notified of changes

Use cases:

- Couples deciding on home purchase
- Client sharing with mortgage broker
- Financial advisor reviewing with client

Technical: WebSocket or polling for real-time sync
 Operational Transform for concurrent edits

3.5.8 Output Versioning & History

```
interface CalculationHistory {
  id: string;
  userId: string;
  calculationType: string;
  inputs: object;
  outputs: object;
  label: string; // User-set: "Option A - Riverside Drive"
  tags: string[]; // ["primary home", "30yr", "high rate"]
  createdAt: Date;
  updatedAt: Date;
  shareToken?: string;
  downloadHistory: DownloadEvent[];
  shareHistory: ShareEvent[];
  isArchived: boolean;
}

// Dashboard shows timeline of all calculations
// User can label each: "House on Oak Street" vs "Condo Downtown"
// Compare any two saved calculations
// Track how rates changed your calculations over time
```

3.5.9 Output Action Toolbar Component (Full Spec)

```
// Component: <OutputActionToolbar />
// Appears: Sticky at bottom of results panel on mobile
//           Horizontal bar above amortization table on desktop

interface OutputAction {
  id: string;
  label: string;
  icon: string;
  tier: "free" | "pro"; // Pro actions show lock icon
  action: () => void;
  shortcut?: string; // Keyboard shortcut
}

const outputActions: OutputAction[] = [
  // DOWNLOAD GROUP
  { id: "pdf-quick", label: "Quick PDF", icon: "📄", tier: "free" },
  { id: "pdf-full", label: "Full Report", icon: "📄", tier: "free" },
  { id: "excel", label: "Excel", icon: "📊", tier: "free" },
  { id: "csv", label: "CSV Data", icon: "📋", tier: "free" },
  { id: "png-card", label: "Image Card", icon: "🖼️", tier: "free" },
  {
    id: "print",
    label: "Print",
    icon: "🖨️",
    tier: "free",
    shortcut: "Ctrl+P",
  }
]
```

```

},

// SHARE GROUP
{ id: "share-link", label: "Copy Link", icon: "🔗", tier: "free" },
{ id: "qr-code", label: "QR Code", icon: "📱", tier: "free" },
{ id: "email", label: "Email", icon: "✉️", tier: "free" },
{ id: "embed", label: "Embed Code", icon: "</>", tier: "free" },

// SOCIAL GROUP
{ id: "twitter", label: "Twitter/X", icon: "🐦", tier: "free" },
{ id: "linkedin", label: "LinkedIn", icon: "💼", tier: "free" },
{ id: "whatsapp", label: "WhatsApp", icon: "💬", tier: "free" },
{ id: "facebook", label: "Facebook", icon: "📘", tier: "free" },
{ id: "telegram", label: "Telegram", icon: "✈️", tier: "free" },
{ id: "reddit", label: "Reddit", icon: "🟠", tier: "free" },

// PREMIUM GROUP
{ id: "save", label: "Save to Account", icon: "💾", tier: "free" },
{ id: "collaborate", label: "Share & Edit", icon: "👥", tier: "pro" },
{ id: "password", label: "Password Protect", icon: "🔒", tier: "pro" },
{ id: "custom-brand", label: "Custom Logo PDF", icon: "🖨️", tier: "pro"
},
{ id: "permanent", label: "Permanent Link", icon: "∞", tier: "pro" },
{ id: "bulk-export", label: "Bulk Export", icon: "📦", tier: "pro" },
];

```

3.5.10 Output Viewing Experience

Results View Modes

Three ways to VIEW results (before downloading):

1. STANDARD VIEW (Default)

Summary cards → Charts → Amortization table
 All on one scrollable page
 Action toolbar sticky on desktop

2. PRESENTATION MODE

Full-screen, distraction-free
 Keyboard arrow navigation between sections:
 → Summary → Charts → Yearly → Monthly
 Great for meetings with advisors
 URL: /calculators/mortgage/basic/#presentation
 Toggle: [🖨️ Presentation Mode] button

3. PRINT PREVIEW MODE

Shows exact what will be printed/PDF'd

Page break indicators visible
Edit options still available
[Exit Preview] [Print Now] [Save as PDF]

Interactive Chart Features in Output

All charts in output panel:

- ✓ Hover tooltips with exact values
- ✓ Click to highlight/isolate data series
- ✓ Zoom in on specific year range
- ✓ Toggle between monthly/yearly view
- ✓ Download individual chart as PNG
- ✓ Accessible data table toggle
(for screen readers and data export)
- ✓ Responsive (redraw on panel resize)

Chart Types:

1. Donut: Monthly payment breakdown (P&I, Tax, Insurance, HOA, PMI)
2. Stacked Bar: Monthly P&I split over loan life
3. Line: Remaining balance decay curve
4. Area: Cumulative interest vs principal paid
5. Milestone markers: 25%, 50%, 75% equity points

3.6 Country-Specific Calculator Variations

United States

Loan Types: Conventional, FHA, VA, USDA, Jumbo

Special Inputs:

- Credit score (FICO)
- Down payment % (triggers PMI < 20%)
- Property type (primary/investment/vacation)
- State property tax rates (auto-populated)
- State income tax (for tax deduction calc)
- Mortgage interest deduction calculator
- FHA MIP (upfront 1.75% + annual 0.55-1.05%)
- VA funding fee (1.4-3.6%)
- USDA guarantee fee

United Kingdom

Loan Types: Repayment, Interest-Only, Tracker, Fixed, Help-to-Buy

Special Inputs:

- Stamp Duty Land Tax (SDLT) calculator built-in
- First-time buyer status (SDLT relief)

- Help-to-Buy Equity Loan %
- Shared Ownership calculator
- Stress test rate (+3% BOE stress test)
- Arrangement fees
- ERC (Early Repayment Charge) calculator
- APRC display

Canada

Special Inputs:

- CMHC Insurance premium
- Provincial Land Transfer Tax
- Stress test qualifying rate
- First Home Savings Account (FHSA)
- First-Time Home Buyer Incentive

Australia

Special Inputs:

- Stamp Duty by state
- First Home Owner Grant by state
- LMI (Lenders Mortgage Insurance)
- APRA buffer rate (+3%)
- Negative gearing calculation
- Comparison Rate display (mandatory)
- Offset account impact

Germany

Special Inputs:

- Grunderwerbsteuer (3.5-6.5%)
- Notarkosten (~1.5%)
- KfW subsidy programs
- Tilgungssatz
- Sondertilgung option

India

Special Inputs:

- EMI calculation
- Section 80C deduction (max ₹1.5L)
- Section 24(b) deduction (max ₹2L)
- PMAY subsidy

- CIBIL credit score
- Pre-EMI vs full EMI option



Special Inputs:

- Expatriate vs UAE national LTV
- Central Bank DBR calculator
- DLD fee (4%)
- Islamic finance options



Islamic Finance (Global)

Products:

- Murabaha (cost-plus-profit)
- Diminishing Musharakah
- Ijara (lease-to-own)
- Istisna (construction)

No compound interest calculations

True ownership transfer tracking

3.7 Additional Calculator Specifications

3.7.1 Affordability Calculator

Inputs: Annual income, monthly debts, down payment, credit score

Outputs:

- Recommended home price range
- Maximum home price
- DTI ratio gauge
- Stress test result

All output/download/share features apply

3.7.2 Refinance Calculator

Inputs: Current balance, current rate, current payment,
remaining term, new rate, new term, closing costs

Outputs:

- New monthly payment
- Monthly savings
- Break-even timeline chart
- Lifetime savings

- Side-by-side comparison table

All output/download/share features apply

3.7.3 Loan Comparison Tool

Compare up to 3 loan scenarios simultaneously
All three exportable in single PDF/Excel
Side-by-side social share card
"Best Pick" AI recommendation included in export

3.8 UI/UX Requirements

3.8.1 Design System

Colors:

Primary: #2563EB (Trust Blue)
Secondary: #10B981 (Financial Green)
Warning: #F59E0B (Caution Amber)
Danger: #EF4444 (Alert Red)
Neutral: #F8FAFC → #0F172A

Typography:

Headings: Inter (700, 600)
Body: Inter (400, 500)
Numbers: JetBrains Mono (financial figures)

Spacing: 4px grid system

Border radius: 8px (cards), 4px (inputs)

3.8.2 Layout Principles

Desktop (>1024px):
[Input Panel Left 40%] | [Results Panel Right 60%]
Real-time calculation updates
Sticky Output Action Toolbar in results panel

Tablet (768-1024px):
Full-width stacked
Floating Share/Download FAB button

Mobile (<768px):
Input → Calculate → Results
Bottom Sheet: Download/Share options
Swipeable chart carousel
"Share" as primary CTA (prominent)

3.8.3 Output-Specific UX Patterns

- ✓ "Results Ready" toast notification after calculation
- ✓ Smooth scroll to results section
- ✓ Highlight key numbers with subtle animation
- ✓ One-click copy for individual values
- ✓ Long-press on mobile for quick share
- ✓ Keyboard shortcut: Ctrl+D = Download PDF
- ✓ Keyboard shortcut: Ctrl+S = Save to account
- ✓ Keyboard shortcut: Ctrl+E = Email results
- ✓ Keyboard shortcut: Ctrl+P = Print
- ✓ Results persist on page refresh (localStorage)
- ✓ "Last calculated: X minutes ago" indicator
- ✓ Warning banner if inputs changed since last download

3.9 SEO Architecture

3.9.1 On-Page SEO Requirements

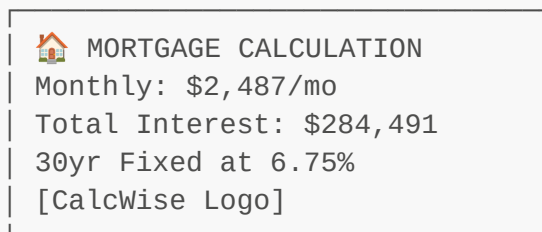
- Per Calculator Page:
- ✓ Unique H1 with primary keyword
 - ✓ 800-1200 word supporting content
 - ✓ FAQ section (5-10 questions with schema markup)
 - ✓ "How to use" guide
 - ✓ Related calculators internal links
 - ✓ Canonical URLs
 - ✓ Open Graph tags (dynamic OG image per calculation)
 - ✓ Twitter Card tags
 - ✓ Schema: SoftwareApplication, FAQPage, BreadcrumbList

3.9.2 Dynamic OG Images (SEO + Sharing Superpower)

UNIQUE FEATURE: Auto-generated Open Graph images per shareable link, showing actual numbers

When user shares `calcwise.com/share/Xk7mP2nQr5:`

→ Facebook/LinkedIn/Slack unfurl shows:



→ 10x more clicks than plain URL share

→ Implemented via `@vercel/og`

3.9.3 Programmatic SEO Pages

Pattern: `/mortgage-calculator/[country]/[state-city]/`

Target: 10,000+ indexed pages within 12 months

Each page: Localized data + 500 words unique content

3.10 Monetization Requirements

3.10.1 Google AdSense Placements

Calculator Pages:

- Below inputs (300x250) – never blocks results
- Between charts and amortization table (728x90)
- Right rail desktop (300x600)
- Below action toolbar (300x250)
- After full amortization table (336x280)

Download/Share Flow:

- Pre-download interstitial (3-second countdown)
"Your PDF is generating... [Ad]"
(Monetizes download action without blocking)
- Post-share confirmation page (728x90)

⚠ Critical Rules:

- NEVER cover input fields
- NEVER cover output results
- NEVER cover download/share buttons
- No ads in print view or PDF
- No ads during active presentation mode

3.10.2 Affiliate Revenue

Contextual CTAs after output:

"Today's Best Rates from Our Partners"

- After mortgage calc: Lenders (Rocket, Better, LendingTree)
- After affordability: Pre-approval CTAs
- After all calcs: Insurance quotes
- All calcs: Credit score check CTA

Revenue model:

- Per lead: \$10-50
- Per funded loan: \$500-2,000

3.10.3 Premium Subscription

Free:

- ✓ All calculators
- ✓ PDF & Excel download (with CalcWise branding)
- ✓ Basic shareable links (7-day expiry)
- ✓ QR code generation
- ✓ Social sharing
- ✓ Email results (to self only)
- ✓ 3 saved calculations
- ✓ Standard print view

Pro (\$9/month):

- ✓ All Free features
- ✓ Permanent shareable links

- ✓ Password-protected links
- ✓ Custom name/logo on PDF (white-label)
- ✓ Email to multiple recipients
- ✓ Unlimited saved calculations
- ✓ Collaborative share & edit
- ✓ Rate alerts (email + SMS)
- ✓ Bulk download (all calculations at once)
- ✓ Advanced Excel (embedded charts)
- ✓ Branded social share cards
- ✓ Priority support

Business (\$29/month):

- ✓ All Pro features
- ✓ White-label embeds (your domain)
- ✓ API access
- ✓ Client management dashboard
- ✓ Bulk client report generation
- ✓ Custom branding on all outputs
- ✓ Team member accounts (up to 5)
- ✓ Webhook notifications
- ✓ CRM integration (Zapier/HubSpot)

3.11 Performance Requirements

Metric	Target	Priority
Core Web Vitals LCP	< 2.5s	Critical
Core Web Vitals CLS	< 0.1	Critical
Core Web Vitals FID	< 100ms	Critical
Calculator Response	< 100ms	Critical
PDF Generation (Quick)	< 2s	High
PDF Generation (Full)	< 8s	High
Excel Generation	< 3s	High
Share Link Creation	< 500ms	High
Image Generation	< 2s	High
QR Code Generation	< 200ms	High
Email Delivery	< 30s	Medium
Page Load (3G)	< 2s	High
Lighthouse Score	> 90	High
Uptime SLA	99.9%	Critical

3.12 Analytics & Tracking

```
events: {  
  // Calculator core  
  'calculation_performed': { calc_type, loan_amount, country },  
  
  // Download events  
  'download_initiated': { format, calc_type, mode },  
  'download_completed': { format, file_size_kb, duration_ms },  
  'download_failed': { format, error_reason },  
  'pdf_options_opened': { calc_type },  
  'print_triggered': { calc_type },  
  
  // Share events  
  'share_link_created': { calc_type, user_type },  
  'share_link_copied': { calc_type },  
  'share_link_opened': { token, is_owner },  
  'qr_code_generated': { calc_type, size },  
  'qr_code_downloaded': { format },  
  'email_sent': { recipient_count, attachments },  
  'email_failed': { reason },  
  
  // Social share events  
  'social_share_clicked': { platform, calc_type },  
  'social_share_completed': { platform },  
  'embed_code_copied': { calc_type },  
  
  // Collaboration  
  'collaborative_share_created': {},  
  'collaborative_edit_made': { editor_type },  
  
  // Premium conversion funnel  
  'pro_feature_blocked': { feature, calc_type },  
  'upgrade_modal_shown': { trigger_feature },  
  'upgrade_clicked': { plan },  
  'upgrade_completed': { plan, revenue }  
}
```

3.13 Security & Compliance

Share Links:

- ✓ Tokens are cryptographically random (not sequential)
- ✓ Rate limiting: 10 share links per hour per IP (guest)
- ✓ No PII stored in share link data
- ✓ Password-protected links: bcrypt hashed
- ✓ Expired links return 410 Gone (not 404)
- ✓ Abuse reporting on share pages

Downloads:

- ✓ No server-side file storage for guest users (generated client-side, never touches server)
- ✓ Authenticated users: S3 with signed URLs (15min expiry)
- ✓ Virus scan on any server-side generated files
- ✓ Rate limiting: 20 downloads per hour per IP

Email:

- ✓ SPF, DKIM, DMARC configured
- ✓ Rate limiting: 5 emails per hour per IP
- ✓ Unsubscribe mechanism in every email
- ✓ GDPR: explicit consent before email capture
- ✓ Email verification before delivery
- ✓ No email address stored without consent

General:

- ✓ HTTPS everywhere (HSTS)
- ✓ CSP headers
- ✓ Input sanitization
- ✓ GDPR + CCPA compliant
- ✓ Cookie consent management
- ✓ MFA support for accounts

3.14 Accessibility (Output & Share Features)

Downloads:

- ✓ PDF tagged for screen readers (PDF/UA standard)
- ✓ Excel includes accessible table structure
- ✓ Color-blind friendly chart palettes
- ✓ Data tables available as alternative to charts

Sharing:

- ✓ Share modal keyboard navigable
- ✓ Copy link button has ARIA label
- ✓ QR code has text alternative
- ✓ Social buttons announce platform name
- ✓ Focus returns to trigger button on modal close

Charts:

- ✓ All charts have data table toggle
- ✓ ARIA roles on chart containers
- ✓ Color + pattern differentiation (not color alone)

3.15 Internationalization (i18n)

Phase 1 (Launch): English only, multi-country logic

Phase 2 (Month 6): Spanish, French, German, Hindi, Arabic

Phase 3 (Month 12): Portuguese, Japanese, Chinese, Dutch

Download localization:

- ✓ PDF in user's language
- ✓ Currency formatting per locale
- ✓ Date formatting per locale
- ✓ Number formatting per locale
- ✓ Decimal separator (. vs ,)
- ✓ RTL PDF support for Arabic (Phase 2)

PART 4: DEVELOPMENT PHASES & TIMELINE

Phase 1: MVP (Months 1-3)

- ✓ Core mortgage calculator (US focus)
 - ✓ Affordability + Refinance calculators
 - ✓ Auto + Personal loan calculators
 - ✓ Basic UK, Canada, Australia variations
 - ✓ OUTPUT SYSTEM:
 - ✓ Quick PDF download
 - ✓ CSV download
 - ✓ Basic shareable links (7-day)
 - ✓ Email to self
 - ✓ Twitter + WhatsApp share
 - ✓ Print view
 - ✓ PNG image download
 - ✓ Google AdSense integration
 - ✓ SEO fundamentals (20 landing pages)
 - ✓ Mobile responsive
 - ✓ GA4 + basic analytics events
- Budget: \$15,000-25,000 dev | 3-4 months solo

Phase 2: Growth (Months 4-6)

- ✓ All country-specific calculators (8 countries)
- ✓ Loan comparison tool
- ✓ Rate data integration
- ✓ User accounts + saved calculations
- ✓ Rate alert system
- ✓ Embeddable widget system
- ✓ OUTPUT SYSTEM (Phase 2):
 - ✓ Full server-rendered PDF (Puppeteer)
 - ✓ Excel with charts
 - ✓ QR code generation + download
 - ✓ All social platforms (LinkedIn, Facebook, Reddit)
 - ✓ Email to multiple recipients with attachments
 - ✓ Dynamic OG images for share links

- ✓ Presentation mode
 - ✓ Permanent links (Pro users)
 - ✓ Password-protected links (Pro)
 - ✓ Programmatic SEO (5,000 pages)
 - ✓ Affiliate integration
 - ✓ Islamic finance calculators
 - ✓ Blog/content platform
- Budget: \$10,000-15,000 additional

Phase 3: Moat (Months 7-12)

- ✓ AI Scenario Engine
 - ✓ Financial Health Score
 - ✓ Premium subscription full launch
 - ✓ OUTPUT SYSTEM (Phase 3):
 - ✓ Collaborative share & edit
 - ✓ White-label PDF (Business tier)
 - ✓ Bulk export
 - ✓ API for report generation
 - ✓ CRM integrations (Zapier, HubSpot)
 - ✓ Calculation history dashboard
 - ✓ Version comparison (how has rate changed?)
 - ✓ API for developers
 - ✓ Multi-language (ES, FR, HI, AR)
 - ✓ Mobile app (React Native)
 - ✓ Programmatic SEO (10,000+ pages)
- Budget: \$20,000-30,000 additional

PART 5: SUCCESS METRICS

Traffic KPIs (12-Month Targets)

Month 3: 10,000 monthly organic visitors
Month 6: 50,000 monthly organic visitors
Month 9: 150,000 monthly organic visitors
Month 12: 400,000 monthly organic visitors

Output & Sharing KPIs

Downloads per visitor:	> 15%
Share link creation rate:	> 8%
Social share rate:	> 5%
Email delivery rate:	> 3%
Shared link click-through:	> 40%

PDF completion rate: > 90%
Download-to-registration: > 12%
(Users who download then sign up)

Revenue KPIs

Month 3: \$200-500/month (AdSense only)
Month 6: \$2,000-5,000/month (AdSense + Affiliate)
Month 9: \$8,000-15,000/month (All channels)
Month 12: \$20,000-40,000/month (All + Premium)

Engagement KPIs

Bounce Rate: < 45%
Avg. Session Duration: > 3.5 minutes
Pages Per Session: > 2.5
Calculation-to-Visit: > 65%
Return Visitor Rate: > 30%
Share-driven Return: > 20% of traffic from shared links

PART 6: LAUNCH CHECKLIST

Pre-Launch:

- ☐ All calculators tested with edge cases
- ☐ PDF output tested across all calculator types
- ☐ Excel output tested in Excel 2016+ and Google Sheets
- ☐ CSV import tested in Excel, Google Sheets, Numbers
- ☐ PNG export resolution verified (retina displays)
- ☐ Print view tested in Chrome, Firefox, Safari
- ☐ Share links tested (create, open, expire)
- ☐ QR codes tested (scan with iOS + Android)
- ☐ Email delivery tested (Gmail, Outlook, Apple Mail)
- ☐ Social share previews verified (OG images)
- ☐ All social share text pre-tested for character limits
- ☐ Download rate limiting tested
- ☐ Share link security (expiry, password protection)
- ☐ Cross-browser: Chrome, Firefox, Safari, Edge
- ☐ Mobile: iOS Safari, Android Chrome
- ☐ Google Search Console configured
- ☐ Sitemap.xml submitted
- ☐ AdSense account approved
- ☐ Ad placements don't cover calculator or outputs
- ☐ Privacy Policy covers share + email data
- ☐ Cookie consent implemented

- ☐ SSL certificate active
- ☐ Page speed > 90 on mobile
- ☐ Core Web Vitals all green
- ☐ GDPR compliance for EU email capture
- ☐ Unsubscribe link in all emails

Post-Launch (Week 1):

- ☐ Monitor download success rates
- ☐ Monitor share link creation
- ☐ Check email delivery rates
- ☐ Monitor PDF generation errors
- ☐ Verify social share click-through rates
- ☐ Check GA4 download/share events firing
- ☐ Monitor AdSense policy compliance
- ☐ Submit to Product Hunt
- ☐ Outreach to financial blogs for widget adoption
- ☐ Begin link-building campaign

PRD Version 2.0 — Living Document Next review: After Phase 1 launch Owner: Product Team Last Updated: 2025